Algebra 10 — Algebraic fractions

Calculator allowed. Marks in brackets. ★ = harder question. Factorise first, every time.

1 Simplify [4]

a $\dfrac{x^2-3x}{x-3}$ **b** $\dfrac{x^2-4}{3x+6}$ **c** $\dfrac{x^2+7x+10}{x^2+3x-10}$

a $\dfrac{x(x-3)}{x-3}=x$. **b** $\dfrac{(x-2)(x+2)}{3(x+2)}=\dfrac{x-2}{3}$. **c** $\dfrac{(x+5)(x+2)}{(x+5)(x-2)}=\dfrac{x+2}{x-2}$.

2 Express as a single fraction in its simplest form [6]

a $\dfrac{3(x+2)}{4}-\dfrac{x-1}{6}$ **b** $\dfrac{x}{x^2+x-6}+\dfrac{2}{x+3}$

a Common denominator 12: $\dfrac{9(x+2)-2(x-1)}{12}=\dfrac{9x+18-2x+2}{12}=\dfrac{7x+20}{12}$.

b Factorise first: $x^2+x-6=(x+3)(x-2)$, so the common denominator is $(x+3)(x-2)$: $\dfrac{x+2(x-2)}{(x+3)(x-2)}=\dfrac{3x-4}{(x+3)(x-2)}$.

3 Simplify [6]

a $\dfrac{4x}{2x+8}\times\dfrac{x^2+6x+8}{x^2}$ **b** $\dfrac{x^2-2x-8}{x+1}\div\dfrac{x+2}{3x^2+3x}$

a $\dfrac{4x}{2(x+4)}\times\dfrac{(x+4)(x+2)}{x^2}$: cancel $(x+4)$, x and 2: $\dfrac{2(x+2)}{x}$.

b Turn the second fraction over and multiply:
$\dfrac{(x-4)(x+2)}{x+1}\times\dfrac{3x(x+1)}{x+2}=3x(x-4)$.

4 Solve [9]

a $\dfrac{2x+1}{3}+x=7$ **b** $\dfrac{3}{x}+\dfrac{x}{x+2}=2$ **c** $\dfrac{x+2}{x-1}-\dfrac{x+1}{x+3}=1$

a Multiply by 3: $2x+1+3x=21\Rightarrow 5x=20\Rightarrow x=4$.

b Multiply by $x(x+2)$: $3(x+2)+x^2=2x(x+2)$, so $3x+6+x^2=2x^2+4x$. Rearranging, $x^2+x-6=0$, so $(x+3)(x-2)=0$: $x=2$ or $x=-3$ (both check: $\frac{3}{2}+\frac{2}{4}=2$ ✓, $-1+3=2$ ✓).

c Multiply by $(x-1)(x+3)$: $(x+2)(x+3)-(x+1)(x-1)=(x-1)(x+3)$. Expand: $x^2+5x+6-(x^2-1)=x^2+2x-3$, so $5x+7=x^2+2x-3$. Then $x^2-3x-10=0$, $(x-5)(x+2)=0$: $x=5$ or $x=-2$.

Total: 25 marks

Check: factorise top and bottom, then cancel whole brackets (never single terms) · add/subtract: common denominator = product of the factors, each used once · divide: flip the second fraction and multiply · equations: multiply every term by the lowest common denominator, then solve · substitute back to check.